SS 2 CHEMISTRY WEEKS 1-6

WEEK: ONE

MEANING OF THE PERIODIC TABLE

The **Periodic Table** is a systematic arrangement of all known chemical elements in order of increasing atomic number, such that elements with similar physical and chemical properties are placed in the same vertical column.

The periodic table is the most important reference chart in chemistry because it summarizes the properties of all elements in a single arrangement

Modern Periodic Law: The Modern Periodic Law states that; the physical and chemical properties of elements are periodic functions of their atomic numbers."

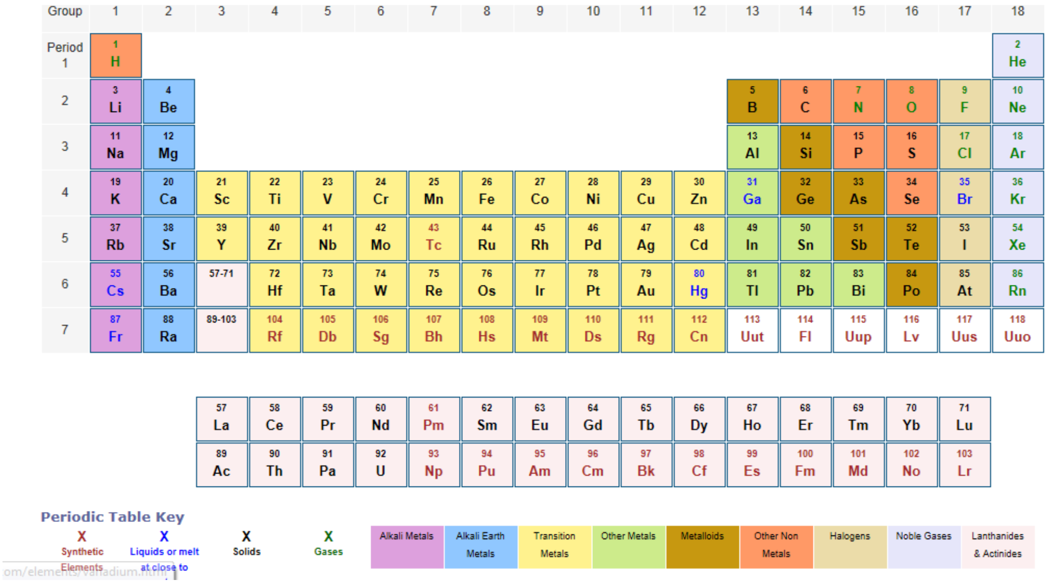
This means that when elements are arranged in order of increasing atomic number, similar properties recur at regular intervals.

The law was proposed by *Henry Moseley* in 1913 after discovering that atomic number, rather than atomic mass, determines the properties of an element.

Explanation

As atomic number increases:

- The electronic configuration changes regularly.
- Elements with the same number of valence electrons appear in the same group.
- Similar chemical properties are repeated periodically.



Group	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Period 1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba	57-71	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra	89-103	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Uut	114 Fl	115 Uup	116 Lv	117 Uus	118 Uuo

57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu
89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr

Importance of the Modern Periodic Law

- Corrected the shortcomings of Mendeleev's periodic table.
- Properly arranged isotopes.
- Removed inconsistencies caused by atomic mass.
- Forms the basis of the modern periodic table.

EVOLUTION OF THE PERIODIC TABLE

Mendeleev's Periodic Table (1869)

Dmitri Mendeleev arranged elements according to increasing atomic mass while grouping similar elements together.

Achievements

- Left spaces for undiscovered elements.
- Predicted the properties of missing elements accurately.
- Grouped similar elements successfully.

Limitations

- Hydrogen had no definite position.
- Isotopes could not be explained.
- Some elements were not arranged in increasing atomic mass.

(d) Moseley's Modern Periodic Table (1913)

Henry Moseley discovered that atomic number is more fundamental than atomic mass. He arranged elements according to increasing atomic number, leading to the modern periodic table.

Advantages

- Correct arrangement of all element, Isotopes occupy the same position.
- Eliminated inconsistencies found in Mendeleev's table, Provides the basis for the modern periodic table.

ARRANGEMENT OF ELEMENTS INTO GROUPS AND PERIODS

Groups

Groups are the *vertical columns* of the periodic table.

The modern periodic table contains *18 groups.*

Characteristics of Elements in the Same Group

- Same number of valence electrons.
- Similar chemical properties.
- Similar valency.
- Undergo similar chemical reactions.

Examples

Group 1 – Alkali Metals: Lithium (Li), Sodium (Na), Potassium (K)

Group 2 – Alkaline Earth Metals: Beryllium (Be), Magnesium (Mg), Calcium (Ca)

Group 17 – Halogens: Fluorine (F), Chlorine (Cl), Bromine (Br), Iodine (I)

Group 18 – Noble Gases: Helium (H), Neon (Ne), Argon (Ar)

Periods

Periods are the horizontal rows in the periodic table.

There *are **7 periods*** in the modern periodic table.

Characteristics of Elements in the Same Period

- They have the same number of electron shells.
- Atomic number increases from left to right.
- Metallic character decreases across the period.
- Non-metallic character increases across the period.

Examples

- Period 1: Hydrogen and Helium.
- Period 2: Lithium to Neon.
- Period 3:
- Sodium to Argon

BLOCKS OF ELEMENTS IN THE PERIODIC TABLES

blocks of elements in the periodic table refer to the way elements are grouped according to the type of subshell (orbital) into which their last (valence) electron enters.

There are four blocks:

1. s-Block Elements

- Location: Groups 1 and 2 (left side of the periodic table), plus hydrogen and helium based on electron configuration.

- Valence electrons: 1 or 2 and the Last electron enters the s-orbital.
- Properties:
- Highly reactive metals (except helium).
- Good conductors of heat and electricity.
- Easily lose electrons to form positive ions.
- Examples: Hydrogen (electron configuration basis), Lithium, Sodium, Magnesium.

2. p-Block Elements

- Location: Groups 13–18 (right side of the periodic table).
- Valence electrons: 3 to 8 and the Last electron enters: the p-orbital.
- **Properties:**
- Contains metals, non-metals, and metalloids.
- Includes many important elements for life.
- Reactivity varies widely.
- **Examples:** Carbon, Nitrogen, Oxygen, Chlorine

3. d-Block Elements (Transition Elements)

- **Location:** Groups 3–12 (middle of the periodic table).
- **Last electron enters:** the **d-orbital**.
- **Properties:**
- Mostly hard, strong metals.
- Good conductors of heat and electricity.
- Often have variable oxidation states.
- Many form colored compounds and act as catalysts.
- Examples: Iron, Copper, Zinc.

4. f-Block Elements (Inner Transition Elements)

- Location: The two rows placed below the main body of the periodic table.
- Last electron enters: the f-orbital.
- Series:
- Lanthanides
- Actinides
- Properties:

- Most are metals.
- Many actinides are radioactive.
- Used in nuclear energy, electronics, and high-strength magnets.
- **Examples:** Cerium, Uranium.

Assignment

1. Draw the outline of the modern periodic table and label all the groups and periods.
2. Explain the modern periodic law.
3. State four differences between groups and periods.

WEEK: Two

Properties of the periodic table and trends

The periodic table helps us predict the physical and chemical properties of elements through periodic trends.

1. Ionization Energy (IE) Is the minimum amount of energy required to remove the outermost (most loosely held) electron from one mole of isolated gaseous atoms to form positive ions.

Example: $Na(g) \rightarrow Na^{+}(g) + e^{-}$

Trend Across a Period, Ionization energy *increases.* This is because nuclear charge increases, atomic radius decreases. Electrons are held more strongly by the nucleus.

Trend Down a group: Ionization energy decreases This is because. Atomic size increases, more electron shells are added and Shielding effect increases. Outer electrons are farther from the nucleus.

2. Electron Affinity (EA): is the amount of energy released when an isolated gaseous atom gains an electron to form a negative ion.

Example: $Cl(g) + e^{-} \rightarrow Cl^{-}(g)$

Trend Across a Period: Electron affinity generally *increases* (becomes more negative). Atoms attract additional electrons more strongly due to increased nuclear charge.

Trend Down a Group: Electron affinity generally *decreases* (becomes less negative). Increased atomic size reduces the attraction for incoming electrons.

3. Atomic Size refers to the overall size or volume occupied by an atom. It is commonly measured using the atomic radius.

Trend Across a Period: Atomic size *decreases.* Increasing nuclear charge pulls electrons closer to the nucleus.

Trend Down a Group: Atomic size *increases,* more electron shells are added.

4. Atomic Radius: Is half the distance between the nuclei of two identical atoms bonded together.

Trend Across a Period: Atomic radius *decreases because* Effective nuclear charge increases, pulling electrons closer.

Trend Down a Group: Atomic radius *increases,* Additional energy levels increase the distance between the nucleus and the outermost electron

5.Ionic Size: Is the overall size of an ion after an atom has gained or lost electrons.

Key Points

- Positive ions (*cations*) are *smaller* than their parent atoms because electrons are lost.
- Negative ions (*anions)* are *larger* than their parent atoms because electrons are gained.

Trend Across a Period: Ionic size generally *decreases* among ions of the same charge.

Trend Down a Group: Ionic size *increases* because additional electron shells are present.

6. Ionic Radius: Is the distance from the nucleus to the outermost electron in an ion.

Trend Across a Period: Ionic radius generally *decreases* among similar ions due to increasing nuclear charge.

Trend Down a Group: Ionic radius *increases* because more electron shells are added.

7. Electronegativity: Is the ability of an atom in a chemical bond to attract the shared electrons toward itself.

Trend Across a Period: Electronegativity *increases.*

- Atoms have greater attraction for bonding electrons due to increasing nuclear charge.

Trend Down a Group: Electronegativity *decreases*, Larger atomic size and greater shielding reduce the attraction for bonding electrons.

Summary Table of Periodic Trends

Property	Across a Period	Down a Group
Ionization Energy	Increases	Decreases
Electron Affinity	Increases (more negative)	Decreases
Atomic Size	Decreases	Increases
Atomic Radius	Decreases	Increases
Ionic Size	Generally, decreases	Increases
Ionic Radius	Decreases	Increases
Electronegativity	Increases	Decreases

DIAGONAL RELATIONSHIP

Because metallic character increases down a group and decreases from left to right along a period, there exists a diagonal relationship between the chemical properties of the first member of a group and that of the second member of the next group as in the cases of lithium and magnesium on one hand, and beryllium and aluminum on the other (see the periodic table)

Li Be B C

Na Mg Al Si

Applications of Periodic Trends

- Predicting the chemical reactivity of elements.
- Explaining the formation of ions.
- Understanding the types of chemical bonds formed.
- Comparing the strengths of metallic and non-metallic elements.

Assignment

1. Define the following terms: Ionization energy, Electron affinity, atomic radius, Ionic radius, Electronegativity
2. State the trend of each property across a period and down a group.
3. Explain why: Atomic size decreases across a period, Electronegativity decreases down a group. Ionization energy increases across a period.

WEEK: Three

BASIC CONCEPT: REACTANTS AND PRODUCTS

Rate of Chemical Reaction

The **rate of a chemical reaction** is the change in the amount (or concentration) of reactants or products per unit time. It can also be defined as *the speed at which reactants are converted into products. Unit is* $moldm^{-3}s^{-1}$

$4NH_{3(g)} + O2(g) \longrightarrow 4NO_{(g)} + 6H_2O_{(g)}$

Reactants Products

Expression for Rate of Reaction

The rate of reaction can be expressed as:

Rate = Change in quantity of reactant or product ÷ Time taken

Formula

Rate = ΔQuantity / ΔTime

OR

Rate = ΔConcentration / ΔTime

Calculations on rate of chemical reaction

Example 1

A reaction produces $120cm^3$ of hydrogen gas in 30seconds.

Calculate the rate of reaction.

Solution

Given:

Volume = 120 cm^3

Time = 30 s

Rate = Volume ÷ Time

Rate = 120 ÷ 30

Rate = 4 cm^3/s

Example 2

A piece of zinc loses 18g in 6minutes during a reaction.

Calculate the reaction rate.

Solution

Rate = Mass lost ÷ Time

Rate = 18 ÷ 6

Rate = 3 g/min

Example 3

The concentration of hydrochloric acid decreases from 2.0 mol/dm^3 to 0.8 mol/dm^3 in 4 minutes.

Calculate the average rate of reaction.

Solution

Change in concentration

= 2.0 − 0.8

= 1.2 mol/dm^3

Rate

= 1.2 ÷ 4

Rate = 0.30 $mol/dm^3/min$

Example 4

A reaction produces 300 cm^3 of oxygen gas in 5 minutes.

Calculate the rate in:

(a) cm^3/min

(b) cm^3/s

Solution

(a)

Rate

= 300 ÷ 5

= 60 cm^3/min

(b)

5 minutes = 300 seconds

Rate

$= 300 \div 300$

$= 1\ cm^3/s$

Collision Theory

Collision theory states that a chemical reaction occurs only when the reacting particles collide with sufficient energy (equal to or greater than the activation energy), and the correct orientation. Only collisions that meet these conditions are called *effective or successful collision*

Types of Collisions

1. Ineffective Collision

This is a collision that does not produce a chemical reaction because: the particles possess insufficient energy, or they collide in the wrong orientation.

Example:
Hydrogen and oxygen molecules may collide many times without reacting if the collisions do not have enough energy.

2. Effective collision

An effective collision is one that results in a chemical reaction because the particles possess enough energy, and they collide in the proper orientation. Effective collisions lead to the breaking of old bonds and the formation of new chemical bond

Conditions for Effective Collision

1. Sufficient Energy

The colliding particles must have energy equal to or greater than the *activation energy.*

Activation Energy (Ea)

Activation energy is the *undergo a minimum amount of energy required for reacting particles to successful chemical reaction.* If particles possess less than this energy, no reaction occurs. *Example.* Hydrogen and oxygen require a spark before combining explosively because the spark provides the activation energy.

ASSIGNMENT:

1. (a) State in three short sentences on the main ideas of the collision theory.
2. State how the rate of a chemical reaction is affected by the following factors.

   (a) Concentration (b) surface area

WEEK: Four

FACTORS AFFECTING THE RATE OF CHEMICAL REACTION

The rate at which reaction takes place will be affected by the following factors:

1. Nature of reactant: The chemical nature of the reactants taking part in the reaction influences the rate of reaction. For instance, if iron, zinc and gold metals are placed in different beaker continually Hydro chloric acid, there will be rapid evolution in the beaker containing the acid and zinc metal wherein the beaker containing Iron metal and the acid there will be slower evolution of hydrogen gas and in the beaker containing gold metal and the acid solution there will be no reaction.
2. Effect of concentration / pressure (gases) of reactants: If reactant particles are crowded in a particular place, their frequency of collision will be faster than if the particles are far to one another. Thus, the more the concentration of reactant particles and the higher the rate of reaction.

a. Pressure affects the concentration of gaseous reactant, the higher the pressure of gaseous reactants, the higher the frequency of collision of the particles the higher the rate of reaction.

3. Effect of surface area of the reactants: the more exposed the area of contact of reacting particles to each other the faster the rate of reaction. For solid reactants the exposed surface area must be increased by subdividing or breaking the solid into smaller pieces.

i. Where the reactants are gases, liquids or solids dissolved in solution, the thoroughness of mixing is of vital importance so as to ensure maximum contact between the reactants.

4. Effect of temperature of reaction mixture: The higher the temperature the higher the rate of reaction of most reactions. When the temperature of a chemical reaction is increased, the number of particles with energies equal to or greater than the activation energy increases. Also, the kinetic energy of the reactant

particles increases which implies increase in frequency of effective collision and increase in the rate of reaction.

5. Effect of the presence of light: Reactions whose rates are affected by light are called photochemical reactions. The reactant energy. They are thus able to overcome the activation energy barrier and react rapidly by a chain reaction.
6. The rate Examples: i. The decomposition of hydrogen peroxide
   ii. The reactions between methane and chlorine
   ii. Photosynthesis
   iii. The conversion of silver halides to grey metallic silver

6. Effect of catalyst: catalyst will alter the rate of a chemical reaction but

Itself does not undergo any permanent change at the end of the reaction.

Catalysts that speed up the rate of a chemical reaction are called positive catalyst. They lower the activation energies of the reactant particles so more reactant particles are able to collide effectively to produce more products. E.g. Manganese (iv) oxide, MnO_2 catalysts is used in the production of oxygen thermal decomposition of $KClO_3$

ENDOTHERMIC AND EXOTHERMIC REACTION

An endothermic reaction is a chemical reaction that absorbs heat energy from its surroundings, as heat is absorbed, the surroundings become cooler, the products contain more energy than the reactants, The enthalpy change ***(ΔH) is positive (+).***

General Equation: Reactants + Heat → Products

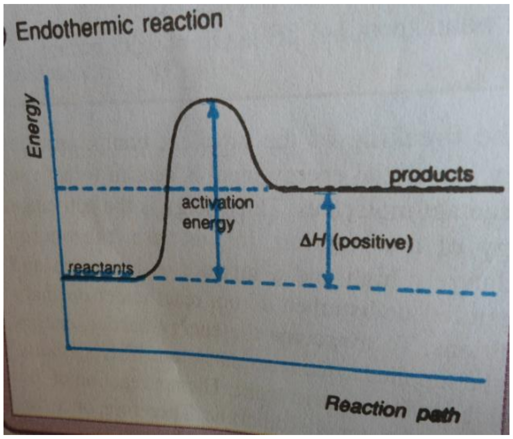
Characteristics of Endothermic Reactions: Heat is absorbed, Temperature of the surroundings decreases, Products have higher energy than reactants, ΔH is positive.

Examples of Endothermic Reactions: Photosynthesis, thermal decomposition of calcium carbonate, Electrolysis of water, Melting of ice. Evaporation of water

Uses of Endothermic Reactions

- Instant cold packs for sports injuries
- Photosynthesis in plants
- Cooling systems
- Industrial decomposition of limestone

(a) $N_2(g) + O_2(g) \longrightarrow 2NO(g)$



ΔH= -180.6KJmol $^{-1}$

Exothermic Reactions

An **exothermic reaction** is a chemical reaction that **releases heat energy into the surroundings**. As heat is released: The surroundings become warmer. The products contain less energy than the reactants. The enthalpy change *(ΔH) is negative (−).*

General Equation: Reactants → Products + Heat

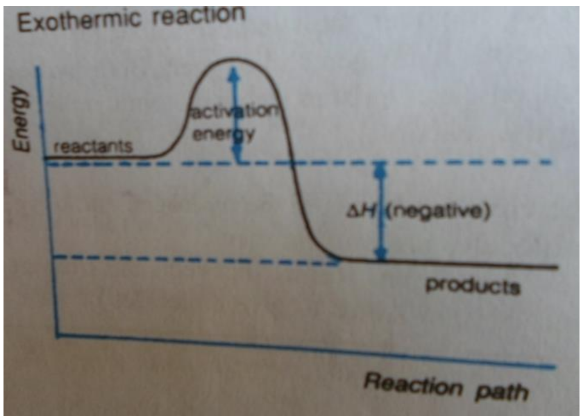
Examples of Exothermic Reactions: Combustion (burning of fuels), Respiration, Neutralization of acids and bases, Condensation of steam

Uses of Exothermic Reactions: Cooking with gas, Heating homes, Power generation, Hand warmers, Automobile engines

(B) Example of exothermic reaction.

$$C(s) + O_{2(g)} \longrightarrow CO_2(g)$$

$$\Delta H = -\ 408 KJmol^{-1}$$



Note: The negative and the positive signs indicating in the above figures indicate exothermic and endothermic reactions respectively.

ASSIGNMENT:

1. State and explain five factors that affect the rate of a chemical reaction.
2. Using suitable examples, describe how temperature and surface area influence the rate of chemical reactions.
3. Differentiate between exothermic and endothermic reaction.
4. Explain the energy changes that occur during endothermic and exothermic reactions using suitable example

WEEK: five

CHEMICAL EQULIBRIUM: A dynamic equilibrium involving a chemical equilibrium. We shall be looking at dynamic V in reversible chemical reactions. A reversible reaction is one that proceeds in both directions under suitable conditions.

Examples are

(a) $N_2O_4 \rightleftharpoons 2NO_2(g)$

(b) $2H_2(g) + O_2(g) \rightleftharpoons 2H_2O(g)$

A reversible reaction is in dynamic equilibrium when both the forward and the backward reactions are occurring at the same rate, thereby producing no net change in the concentrations of the reactants or products

EVALUATION: Write four examples of reversible reactions.

LE CHATELIER'S PRINCIPLE

Le hatelier's principal stares that if an external constraint such as a change to temperature, pressure or concentration is imposed on a chemical system in equilibrium, the equilibrium will shift so as to annul or neutralize the constraint.

FACTORS AFFECTING EQUILIBIUM OF A CHEMICAL REACTION

1. **EFFCT OF A CHANGE IN TEMPERATURE** e.g. For the following systems at equilibrium

   (a) $2SO_3(g) + O_2(g) \rightleftharpoons 2SO_3 \triangle H = - 395.7\ Kjmol^{-1}$

The forward equation is exothermic i.e. it involves increase in temperature whereas the backward equation is endothermic or it involves decrease in temperature.

If the temperature of the system is increased:

Increase in the temperature of the system will cause the equilibrium positon of the backward reaction, that is , reactant formation.

If the temperature of the system is decreased:

Decrease in the temperature of the system will cause the equilibrium position to shift to the right favouring the forward reaction, that is, product formation.

(b) $N2(g) + O_2 \rightleftharpoons 2NO(g)$

$\triangle H = +90.4Kjmol^{-1}$

When the system is at equilibrium, the forward reaction is endothermic and the backward reaction is exothermic.

If the temperature of the system is increased:

Increase in the temperature of the system will cause the equilibrium position to shift to the right favoring forward reaction. i.e. product formation.

If the temperature of the system is decreased:

Decrease in the temperature of the system will cause the equilibrium position to shift to the left favoring the backward reaction.

2. **EFFECT OF A CHANGE IN PRESSURE:**

   For a change in pressure to affect a chemical system in equilibrium.

   (a) One of the reactants in the or products in the reversible reaction must be gaseous.

   (b) The total number of gaseous molecules on the left side of the equation must be different from the total number of moles of gaseous molecules on the right side.

For the following systems at equilibrium

(a) $3H_2(g) + N_2(g) \rightleftharpoons 2NH_3(g)$

The forward reaction is from high pressure to low pressure and the backward reaction is from low pressure to high pressure.

Increase in pressure of the system:

When the pressure of the system is increased the equilibrium position will shift to the right favoring the forward reaction that is, the product formation.

Decrease in the pressure of the system:

When the pressure of the system is decreased the equilibrium position will shift to the left favoring the backward reaction that is reactant formation.

(b) $N_2O_4(g) \rightleftharpoons 2NO_2(g)$

When the system is at equilibrium, the forward reaction involves an increase in the pressure of the system and the backward reaction involved a decrease in the pressure of the system.

If the pressure of the system is decreased:

A decrease in the pressure of the system will cause the equilibrium position to shift to the right favoring the forward reaction that is, product formation.

If the pressure of the system is increased:

An increase in the pressure of the system will cause the equilibrium position to shift to the left favoring the backward reaction that is, reactant formation

(c) $H_2(g) + I_2(g) \longrightarrow 2HI(g)$

A change in pressure will not affect this system at equilibrium because the number of moles of the reactants is the same as the number of moles of the product.

(d) $3Fe(s) + 4H_2O(g) \rightleftharpoons Fe_3O_4(s) + 4H_2(g)$

A change in pressure will not affect this system at equilibrium because the number of moles of gaseous reactant and product is the same.

3. EFFECT OF A CHANGE IN CONCENTRAION

E.g. $3Fe(s) + 4H_2O\ (g) \rightleftharpoons Fe_3O4(s) + 4H_2(g)$

If the concentration of any of the reactants is increased the equilibrium position will shift to the right but if it is decreased it will shift to the left.

Also, removal of any of the reactant or product will also cause the equilibrium position to shift. E.g. removal of hydrogen gas from the system will cause the equilibrium position to shift to the right.

4. **EFFECT OF A CATALYST**

Addition of a catalyst to a system in equilibrium will not affect the equilibrium position. Instead, addition of catalysts will only make equilibrium state to be reached attained faster.

EXPRESSION OF EQUILIBRIUM CONSTANT K_C

In a reaction: $aA + bB \rightleftharpoons cC + dD$

The equilibrium constant K_C is expressed as follow:

$$K_C = \frac{[C]^c D^d}{[A]^a [B]^b}$$

Note: The species must be written in a square bracket as shown above e.g. in the reaction:

(a) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

The equilibrium expression

$$K_C = \frac{[SO3]^2}{[SO2]^2 [O_2]}$$

(b) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

$$K_C = \frac{[NH3]^2}{[N2]^2 [H_2]}$$

©. $N_2(g) + O_2 \rightleftharpoons 2NO(g)$

$$\frac{[NO]^2}{[N2] [O_2]}$$

ASSIGNMENT

1. Consider the following equation

   $H_2S(g) + M^{2+} \rightleftharpoons MS(aq) + 2H^{+}(aq)$

   State and explain the effect of each of the following can the equilibrium position

   (a) Increase in temperature

   (b) Addition of solution of $M(NO_3)_2$

   (c) Addition of acidified $KmnO_4(aq)$

2. What is the effect of each of the following on the equilibrium position of the system indicated below?

   $A(s) + B(g) \rightleftharpoons C(g) \Delta H = - xKLmol^{-1}$

   (a) Cooling the system

   (b) Removing

   (c) As soon as it is formed

3. Write the express of equilibrium constant for

   $2AB_2(g) + B_2(g) \rightleftharpoons 2AB_3(g)$

4. Consider the reaction represented by the equation

   $N2O_4(g) \rightleftharpoons 2NO_2(g)$

   $\Delta H = +57.2Kjmol^{-1}$

(a) When is the reaction said to be at equilibrium

(b) Mention two conditions that can favour the forward reaction.

(c) Name the principle involved in (b) above

(d) Addition of catalyst.